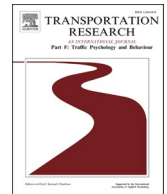




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How knowledge of cycling practice and changes to the British Highway Code relate to perceptions of cyclist discourtesy

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ABSTRACT

In 2022 the UK Highway Code had a major update including advice for cyclists around lane positioning that reflects existing safety training practice and guidance for drivers to leave at least 1.5 m of lateral clearance when overtaking cyclists. This study uses a repeated measures design to examine the impact of changes to the Highway Code on knowledge of cycling road positioning practice, and the effect of this knowledge on perceptions of cyclist discourtesy.

Knowledge of cyclist positioning practice improved significantly between the two surveys. A hierarchical multiple regression found that cyclist positioning knowledge score added significantly to the model predicting perceptions of cyclist discourtesy, with the change in R^2 between the two models being significant. A separate hierarchical multiple regression, with cyclist positioning knowledge as the DV and Highway Code knowledge as the additional stage two IV, was not significant.

The findings showed that while the highway code changes did not predict improvements in knowledge, drivers with better knowledge did have more positive opinions of cyclist behaviour, when controlling for their own participation in cycling.

This was a novel study in applying a repeated measures design before and after the changes to the UK Highway Code. The findings will help inform development of interventions designed to improve road safety for cyclists.

1. Introduction

Driver aggression and risky manoeuvres around cyclists have been shown to be a barrier for many people to travel by bicycle (Davies et al., 1997) and creates a collision risk for those people who do cycle (Aldred 2016). Links have been made between driver rule knowledge and more positive attitudes towards cyclists (Rissel et al., 2002). The current research explores this link further by looking at how knowledge of formal guidance for cyclists could support more positive interpretations of cyclist behaviour by drivers.

The UK Highway Code (Department for Transport, 2022) contains both advice and legally enforceable rules for all users of public roads in the UK. In January 2022 some major changes to the Highway Code were published. This included advice to leave at least 1.5 m of lateral clearance when overtaking cyclists at speeds of up to 30mph, and more space at higher speeds. It also gave more recognition to aspects of training that have been delivered nationally under the Bikeability cyclist training scheme since 2007 (The Bikeability Trust, 2019). The Bikeability lane position training now endorsed by the Highway Code includes never to ride at the very edge of the

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road on the road shoulder or in the gutter. Instead cyclists are to take a Left Lane position at least 0.5 m from the kerb when it is safe for following traffic to overtake, and to ride in a Centre Lane position in slow traffic and when overtaking would be unsafe.

The Highway Code update placed a new expectation on motorists in relation to how they interact with cyclists. A YouGov survey commissioned by Cycling UK (YouGov, 2022) found that 90 % of drivers could recall that there had been some recent changes to the highway code. However this survey also found 66 % of the drivers who were aware changes had been made did not know that cyclists should position themselves in the centre of the lane on quiet roads and in slow-moving traffic. Furthermore 70 % of the drivers who were aware changes had been made did not know that drivers should leave at least 1.5 m of space when overtaking cyclists. This survey required participants to make the correct selection on multiple choice questions, so it did not require participants to recall the rules verbatim. Respondents were encouraged not to guess and had the option to indicate they didn't know the answer. However the multiple choices on offer could have allowed some respondents to make a correct educated guess by selecting the most plausible option, so may slightly overestimate the level of Highway Code knowledge.

The aim of the current study is to evaluate whether these changes in the Highway Code influenced drivers' perceptions of cyclists' road positioning, and whether drivers who do have this knowledge make different interpretations of cyclist behaviour.

1.1. Behaviour towards cyclists

Cycle training schemes The Bikeability Trust (2019), Cycling 4 Safety (2022), The Center for Cycling Education (2021) and the UK Highway Code (Department for Transport, 2022) encourage cycling in a centre lane position in certain scenarios, and at least 0.5 m from the kerb at other times. This guidance is designed to discourage overtaking in situations where to do so would be risky. However, both Rubie et al. (2020) and Mackenzie, Dutschke and Ponte (2021) found that riding further from the kerb was related to smaller passing distances. This demonstrates a conflict between official guidance on safe and responsible cycling and how drivers react to cyclists who are following this official guidance. The experimental results indicating closer passing distances when cycling further from the kerb are supported by Aldred (2016). This study found many riders who were aware of the advice to ride in the centre of their lane were nevertheless unwilling to do so for fear of provoking an angry or impatient response from motorists.

There is a methodological limitation with using lateral passing distance to evaluate the effect of rider positioning, in that it is unable to count potentially risky passes that did not happen because the positioning of the cyclist successfully discouraged the driver from attempting an overtake in the first place. Rasch et al. (2020) provides evidence that riding further from the kerb can prevent overtakes in situations where safety margins are smaller. The study, conducted in Sweden on a test track found that when a robot cyclist was in a centre-lane position, 50 % of drivers waited for an oncoming vehicle to pass before overtaking, compared to 29 % when the robot cyclist was closer to the kerb.

Other studies have examined factors that may influence driver attitudes to cyclists. These studies have not quantified how these factors affect lateral passing distance but do identify factors that could influence drivers' decisions when contemplating a passing manoeuvre where lateral clearance is minimal. For example Basford et al. (2002) found that drivers recognised the need to take care around cyclists but this was somewhat counteracted by perceived social pressure from other drivers. This study also found that infrastructure that failed to give clear ownership of space or prompted improvised driver responses, was associated with drivers perceiving cyclists as less courteous as well as drivers being more likely to execute manoeuvres that could be alarming to cyclists. Heesch et al. (2011) found a range of factors that were associated with cyclists recalling harassment from motorists while riding. These factors comprised being younger, less obese, having more than 2 years' experience cycling, higher cycling frequency, living in socio-economically advantaged areas, cycling for recreation and cycling for competition. Gender was not linked to levels of harassment. These factors conceivably correlate with young, fit cycling enthusiasts attracting more harassment than people cycling for utility purposes or out of economic necessity. A possible reason for this could be a perception that cycling for sport is optional while cycling for transport is necessary, so the source of any inconvenience experienced when encountering sport cyclists may be seen as less legitimate than when encountering utility cyclists. Shinar (1998) suggests drivers are more likely to become angry when delayed if the source of the delay is viewed as illegitimate.

1.2. Knowledge, attitude and behaviour

Drivers who also ride bicycles (cyclist-drivers) have been shown to have a more positive attitude towards cyclists than drivers who don't cycle, with in-group bias being discussed as likely to underpin the more positive attitude towards cyclists (Fruhen and Flin, 2015). In addition to in-group bias, more positive attitudes towards cyclists are also associated with higher levels of general road rule knowledge (Rissel et al., 2002). Oldmeadow et al. (2019) gives more insight into this relationship by demonstrating that general road rule knowledge is related to greater perceived legitimacy of cyclists as road users, and these higher levels of perceived legitimacy are associated with lower levels of driver anger. Johnson et al., (2014) found cyclist-drivers were 1.5 times more likely than non-cyclists to report safe driving behaviours when overtaking cyclists, had better road rule knowledge, were less likely to report frustration when wanting to overtake a cyclist and were less likely to report that cyclists are unpredictable. This unpredictability of cyclists has been found to be a source of irritation among drivers (Basford et al., 2002). These studies indicate that negative attitudes towards cyclists can be mediated by road rule knowledge by making cyclists seem more legitimate and predictable.

The Theory of Planned Behaviour (TPB) (Ajzen, 1991) posits that behavioural intentions are underpinned by beliefs and attitudes towards the behaviour, beliefs around subjective norms of behaviour and beliefs around one's ability to exhibit the behaviour. Negative attitudes towards cyclists have been linked to more aggressive driving behaviour around them (Fruhen and Flin, 2015). The attitude of a driver towards safe overtaking of a cyclist may therefore be influenced by their attitude towards the cyclist. This

conceptual link between attitudes towards cyclists and attitudes towards driving behaviour when encountering them is illustrated in Fig. 1:

1.3. Importance of perspective

It is important to consider whether participant responses can be skewed by the perspective from which stimuli are presented. For example the fundamental attribution error (or self-serving bias) is the tendency for people to blame their own errors on external factors and credit their successes to their own virtues. This was demonstrated in a study that found people who were given the role of a participant in a scenario exhibited self-serving bias while those who were observers did not (Campbell and Sedikides, 1999). Similarly, the actor-observer bias is the tendency to attribute other people's errors to character flaws but one's own errors to external causes. This is shown in a study that found when inconvenienced by another road user, people have a tendency to blame the other road user's behaviour on personality flaws and underestimate the extent to which they were responding to external factors (Wickens, Mann and Wiesensthal, 2013). If a driver is inconvenienced by behaviour interpreted as symptomatic of a character flaw, they are more likely to direct an angry response towards the offending road user than if the exact same behaviour was interpreted as being in line with expected norms. This creates a situation where a driving manoeuvre that poses a risk to a cyclist may be seen as justifiable by the driver, if the cyclist's behaviour is at odds with the expected norm. A lack of knowledge around how cyclists are taught to interact with their environment may increase the likelihood of cyclist's behaviour being interpreted as subverting the expected norms, reducing the driver's inhibition towards the risky manoeuvre.

Knowledge among drivers of recommended cyclist road positioning practice was found to be a mediating factor in perceptions of cyclists' liability in footage of close passes by motor vehicles (Cubbin et al., 2024). The stimuli used in that study were filmed from the perspective of the cyclist experiencing the close pass. Participants were asked questions to measure the extent to which they believed the cyclist bore some responsibility for the close pass. As they were viewing the pass from the perspective of the cyclist, the actor-observer bias suggests survey participants would interpret the situation as though they were the cyclist (actor), rather than as an independent observer or from the perspective of the motor vehicle driver. The current study shifts this perspective to that of a driver approaching a cyclist from behind. This will frame the cyclist as being distinct from the actor in the scenario, meaning any actor-observer or self-serving bias will be aligned with the experience of drivers when they encounter cyclists on the road. This will minimise uncertainty in how the results might apply to drivers in real world situations.

1.4. Synthesis

Infrastructure can give clear signals to drivers on ownership of space that will positively or negatively affect their attitudes towards the cyclists they overtake Basford et al. (2002). However it appears that knowledge of road rules around cycling can also influence these encounters (Oldmeadow et al., 2019). Knowledge that allows a driver to understand cyclists' positioning choices, and even their choice to be on the road in the first place, may help overcome the fundamental attribution error – in this case attributing their frustration at being delayed to a personality or behavioural flaw in the cyclist. Furthermore, greater cycling-specific knowledge may support more positive attitudes towards cyclists by helping drivers become better at predicting cyclist behaviour and being more likely to view that behaviour as being consistent with the expected norm.

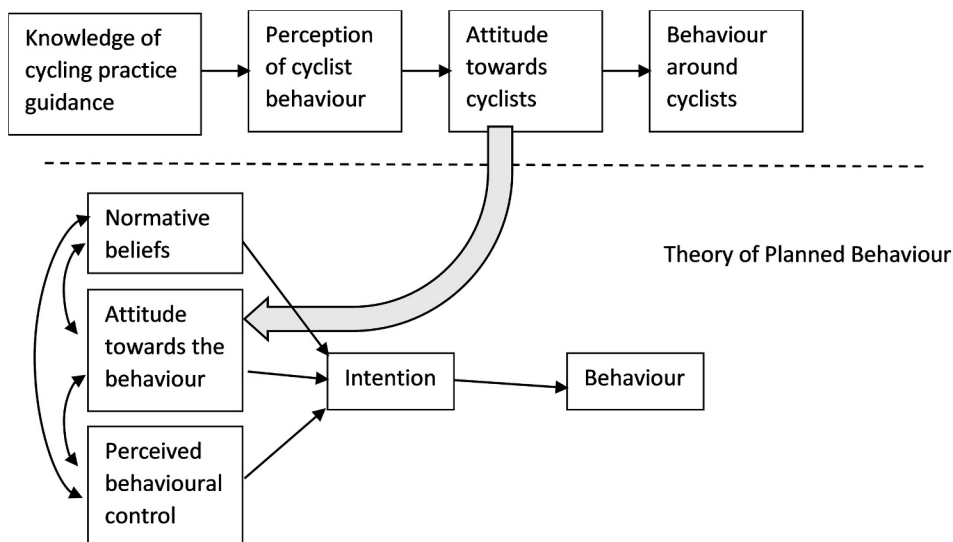


Fig. 1. Logic model linking knowledge of cyclist positioning with behaviour around cyclists, and how it links to the Theory of Planned Behaviour.

1.5. Hypotheses

Existing research suggests improved knowledge among drivers of cyclist positioning is related to increased perceptions of cyclist legitimacy, and this perceived legitimacy could influence greater lateral clearance afforded to cyclists when overtaking. Changes to the UK Highway Code and accompanying publicity may have the potential to increase the level of knowledge among UK drivers of cyclist positioning practice, leading to the following hypotheses:

- H1:** People with a better understanding of cyclist positioning perceive cyclists as being less discourteous above and beyond perceptions related to how much their lifestyle involves driving, level of driving experience, cycling frequency and road user identity.
- H2:** The updates to the Highway Code have improved general knowledge of the most appropriate cyclist positioning in various scenarios.

1.6. Research questions and objectives

Our study builds upon the work of Cubbin et al. (2024) which found that knowledge of recommended cyclist road-positioning among UK drivers was a mediating factor that predicted lower levels of liability apportioned to cyclists in close passes while controlling for Driving Purposes (DP) and road user identity (RUI). DP measures the number of purposes for which a person uses a car, it is not a measure of driving dependence as it does not account for availability or willingness to use alternatives, but it does show the extent to which a participant’s lifestyle involves driving. RUI is a 100 point scale where participants can place themselves, ranging from 0 = “I’m more of a driver” to 100 = “I’m more of a cyclist”.

The following research questions have been formulated to determine if updates to the Highway Code have improved driver understanding of cyclist road positioning and if variations in this level of understanding are related to general perceptions of cyclist discourtesy among drivers:

- Research question 1:** Does greater knowledge of cyclist positioning significantly predict perceptions of cyclist legitimacy above and beyond the predictive ability of DP, driving experience, cycling frequency and RUI?
- Research question 2:** Has the level of knowledge around cyclist positioning improved since the January 2022 Highway Code update, and can any change be attributed to the highway code changes?

The objectives of this research aim to evaluate the impact on driver attitudes to cyclists resulting from changes made in 2022 to the UK Highway Code advice for cyclists. It further aims to determine if drivers who are better able to select appropriate lane positions for cyclists, have more positive attitudes towards cyclists in general. It will analyse the relationship between Highway Code knowledge and any changes in the ability of a within-subjects group of study participants to select appropriate lane positions for cyclists. It will also analyse the relationship between the ability to select appropriate lane positions and general attitudes towards cyclists, while controlling for potential in group and out group biases. This is to determine if improved cycling specific knowledge can predict more positive attitudes to cyclists, and whether the changes to the Highway Code have had any effect on this.

2. Material and methods

2.1. Participants

A survey of UK drivers was carried out in August 2020 to investigate opinions on appropriate cyclist road positioning across a range of road environments and driver characteristics. This survey collected information on which position within the carriageway participants thought was most appropriate for a cyclist to adopt, as well as information on participants’ experience of driving and cycling and their self-reported tendencies for how they react when something makes them angry while driving. Participants were recruited via the Prolific survey platform which pays UK minimum wage pro-rata for time taken to complete surveys.

Table 1
Survey participant demographics.

Survey Participant overview			
	Survey 1 (Aug 2020)	Survey 2 (Dec 2023 – Feb 2024)	Population (GB adult driving licence holders) ¹
Sample size	N = 421	N = 167	N = 42,120,966
Age (years)	(M = 30.88, SD = 13.96)	(M = 39.56, SD = 13.718)	(M = 50.48, SD = 16.98)
Men	(n = 207; 49.2 %)	(n = 77; 46.1 %)	(n = 22,536,533; 53.5 %)
Women	(n = 204; 48.5 %)	(n = 88; 52.7 %)	(n = 19,584,433; 46.5 %)
Non-binary/other/not stated	(n = 10; 2.4 %)	(n = 2; 1.2 %)	not recorded
Rides a bicycle at least once a year	(n = 159; 38 %)	(n = 55; 30 %)	(31 % ²)

1 Department for Transport, 2024.
2 Data for England in 2023 (Department for Transport, 2025).

When the Highway Code was updated in 2022 it created an opportunity to re-run the original questionnaire to study whether participants' beliefs around cyclist road positioning had changed following the Highway Code update. Participants who took part in the August 2020 survey, that were still active users of the Prolific platform in 2023, were invited to complete a second iteration of the survey to allow a repeated measures analysis to be carried out. This approach allowed their responses to be compared before and after changes were made to the Highway Code with its related publicity campaigns and media coverage. The demographics of the participants are summarised in [Table 1](#).

2.2. Stimuli

The survey questions are summarised in [Table 2](#) and included multiple choice questions depicting 13 different scenarios, each showing a picture of a female cyclist from the perspective of a motorist approaching from behind, where the road environment and rider-to-kerb distance were varied. Each scenario was repeated at three different distances to the camera making 39 questions in total. One example is shown in [Fig. 2](#).

The cyclist was wearing casual clothing, a helmet and a yellow hi-vis vest. In each set of three pictures the rider was shown on the same road at the same distance, but variously positioned immediately adjacent to the kerb, around 0.5–1.0 m from the kerb, and in the centre of the lane. The 13 scenarios were chosen to represent situations where a rider should be in a more central position due to road narrowing caused by parked cars or unsafe overtaking conditions caused by bends, as well as scenarios where it would be appropriate for a cyclist to take a left lane position to help a following vehicle to overtake. These scenarios are outlined in [Table 3](#):

The participants were prompted to select which road position they believed to be most appropriate for the cyclist to take, these were either A – “Shoulder” (on the road shoulder or in the gutter), B – “Left Lane” position around 0.5–1.0 m from the kerb or C – “Centre Lane” position (in the centre of the lane or close to the centre of the road for a single lane road). Bikeability training and the Highway Code both teach that position A is never appropriate, with position B being appropriate where it is safe for a following vehicle to overtake and position C being appropriate where the cyclist is travelling close to motor vehicle speeds or where traffic, road layout or other factors mean it is not safe for a following vehicle to overtake.

Bikeability and Highway Code guidance defines position A (shoulder) as never being correct, so any answer giving position A was

Table 2
Survey question set.

Question	August 2020 survey	December 2023 survey
Select the road position recommended by the Highway Code for a cyclist approaching a junction or road narrowing.		Included
a. Around 0.5 m from the left edge of the road		
b. In the middle of their lane		
c. In the centre of the road		
Use the slider to indicate how much clearance in metres should be given when overtaking a cyclist at speeds of up to 30mph, according to the Highway Code		Included
- Clearance in metres 0–5 at 0.5 m increments		
Select the correct answer below, or tick “I don’t know” if you are unsure		Included
a. The Highway Code had its last major update in April 2019		
b. The Highway Code had its last major update in September 2014		
c. The Highway Code had its last major update in June 2008		
d. I don’t know when the Highway Code had its last major update		
What is your gender	Included	Included
What is your age	Included	Included
Do you hold a current drivers’ licence?	Included	Included
How many years have you held your drivers’ licence for?	Included	Included
Click any of the following statements that apply to you		Included
- I drive for part or all of my job		
- I drive to get to my place of work or study		
- I drive for personal business such as shopping		
- I drive to attend social occasions or leisure activities		
- Driving is a leisure activity for me		
How often do you ride a bicycle on the road?	Included	Included
- Never or almost never		
- A few times a year		
- At least once a month		
- At least once a week		
Use the slider to indicate how you see yourself as a road user		Included
- Slider from “I’m more of a driver” to “I’m more of a cyclist”		
You will now be presented with a series of images. For each set of images, please select the most appropriate road position for the cyclist to be in.	Included	Included
Please indicate the extent to which you agree or disagree with the following statements:		Included
- Cyclists’ behaviour is often unpredictable		
- Cyclists have a tendency to be arrogant		
- Cyclists’ behaviour is often irresponsible		
- Cyclists frequently take up space meant for motor vehicles		

Here is a 30 mph, two-lane road in a suburban area. Select the most appropriate road position for the cyclist to be in.

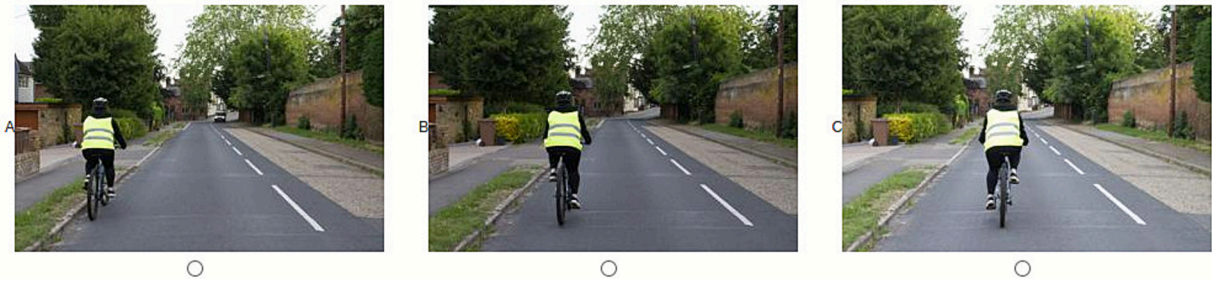


Fig. 2. Example of survey stimuli showing rider at three different lateral road positions.

Table 3
Survey stimuli scenario summary.

	Suburban 20mph	Suburban 30mph	Rural 30mph	Rural 60mph
Two-lane simple		1 x 3 positions		1 x 3 positions
Two lane with parked cars	1 x 2 positions	1 x 2 positions 3 x 3 positions		
Two lane left bend		1 x 3 positions		
Two lane right bend			1 x 3 positions	
Two lane with oncoming traffic				1 x 3 positions
Single lane simple				1 x 3 positions
Single lane left bend				1 x 3 positions
Single lane right bend				1 x 3 positions

given zero points. Guidance was sought from two Bikeability Training Managers who advised that defining the ‘correct’ position involves an element of subjectivity and can also depend on factors that cannot be shown in a photo, such as rider speed. In many situations the Bikeability Training Manager guidance deemed B as slightly better than C (or vice versa), but neither being categorically wrong, so both options were made available. By answering B or C respondents demonstrated the knowledge that A was incorrect so an answer of B or C were both given a score of 1 point.

2.3. Measures and analysis

Research Question 1: “Does greater knowledge of cyclist positioning significantly predict perceptions of cyclist legitimacy above and beyond the predictive ability of DP, driving experience, cycling frequency and RUI?”

To answer this research question a test was formulated for hypothesis #1 (that people with a better understanding of cyclist positioning perceive cyclists as being less discourteous above and beyond perceptions related to how much their lifestyle involves driving, level of driving experience, cycling frequency and road user identity) by applying a bootstrapped hierarchical multiple regression to determine if cycling position knowledge significantly predicts perceptions of cyclist discourtesy. The regression included variables identified in Cubbin et al. (2024) that significantly predicted measures of perceived liability for cyclists in video clips of close-pass overtakes by motor vehicles, they comprise:

Driving Purposes: This is a count of the number of different driving purposes the participant regularly engages in. Participants were asked to select all of the following statements that were applicable to them; ‘I drive for all or part of my job’, ‘I drive to get to my place of work or study’, ‘I drive for personal business such as shopping’, ‘I drive to attend social occasions or leisure activities’, ‘Driving is a leisure activity for me’. The DP score was derived by scoring 1 point for each of the driving purpose types selected. This approach infers the extent to which a respondent’s lifestyle involves driving by focussing on purposes rather than mileage or frequency and was used in Cubbin et al. (2024). The approach was adapted from established practice in using driving purpose categories to measure car dependence from UK Department for Transport National Travel Survey data (Lucas, 2009).

Years licence held: As a proxy measure of driving experience. This is an established measure used in multiple studies included in a meta-analysis of driver experience (Robbins and Chapman, 2019).

Frequency riding bicycle on road: Selected from the following options; Never or almost never, A few times a year, At least once a month, At least once a week.

Road-user self-identity (RUI): As measured on a sliding scale from 0 (100 % driver) to 100 (100 % cyclist).

Research Question 2: “Has the level of knowledge around cyclist positioning improved since the January 2022 Highway Code update, and can any change be attributed to the highway code changes?”

To answer this research question a test was formulated for hypothesis #2 (that the updates to the Highway Code have improved general knowledge of the most appropriate cyclist positioning in various scenarios) by using a t-test to compare changes in applied

knowledge of cyclist positioning and then a bootstrapped hierarchical multiple regression to determine if these changes can be attributed to the Highway Code updates.

For participants who took part in both surveys, a *t*-test was used to compare changes between the two surveys in the proportion of answers indicating they believed that A (tight to the kerb) was the most appropriate position for the cyclist to take. This was to determine if there has been a change in respondents' understanding of cyclist positioning, where a reduction in the proportion of answers stating position A is most appropriate would indicate an improvement in knowledge.

Two distinct measures were made in relation to the Highway Code; Highway Code recollection was measured by answers to questions on when the most recent major update to the Highway Code was made. Highway Code knowledge was measured by accuracy in answering questions on Highway Code advice for cyclist positioning.

A bootstrapped hierarchical multiple regression was carried out to determine if there was a relationship between variables recorded in the survey, including scores on Highway Code knowledge and recollection of the changes, and the level of improvement on the cyclist positioning questions between the two surveys. Fig. 3 shows how the planned analyses relate to the data collected in both stages of the study.

3. Results

The first iteration of the survey conducted in 2020 received 421 complete responses that were included for analysis. Of these 421 respondents, 167 completed the second iteration of the survey in 2023.

Table 4 summarises cyclist positional knowledge scores for all the participants in both surveys. This shows that at least 52.2 % of responses in 2020 showed a lack of understanding of Bikeability advice on cyclist positioning for the scenario depicted. Most of the remaining responses were more conservative in the amount of space they afforded the cyclist by selecting B (left lane) rather than C (centre lane). The subset of participants in survey 1 who also took part in the second survey, allowed even less space to cyclists with over 53 % of responses selecting the A (tight to kerb) position. The same participants showed improved knowledge in the second survey with only 38.7 % selecting the A position and twice as many answers indicating the C position than they did on the first survey.

A paired samples *t*-test showed a significant increase in cyclist position knowledge score between survey 1 and survey 2 among participants who completed both surveys $t(166) = -6.445, p < 0.001$.

3.1. Results for RQ1 "Does greater knowledge of cyclist positioning significantly predict perceptions of cyclist legitimacy above and beyond the predictive ability of DP, driving experience, cycling frequency and RUI?"

A simple exploratory correlation was conducted between the variables measured in the survey using Pearson correlation. The Pearson coefficients in Table 5 show a significant negative correlation between perceptions of cyclist discourtesy and RUI as a cyclist ($r = -0.272, p < 0.01$) and frequency cycling ($r = -0.308, p < 0.01$). There is a significant positive correlation between perceptions of cyclist discourtesy and age ($r = 0.275, p < 0.01$) and years licence held ($r = 0.279, p < 0.01$). Age and years licence held are strongly co-correlated ($r = 0.948, p < 0.01$). There is not a significant correlation between perceptions of cyclist discourtesy and cyclist position scores.

The relationship between scores on cyclist positioning knowledge and perceptions of cyclist discourtesy was tested using a

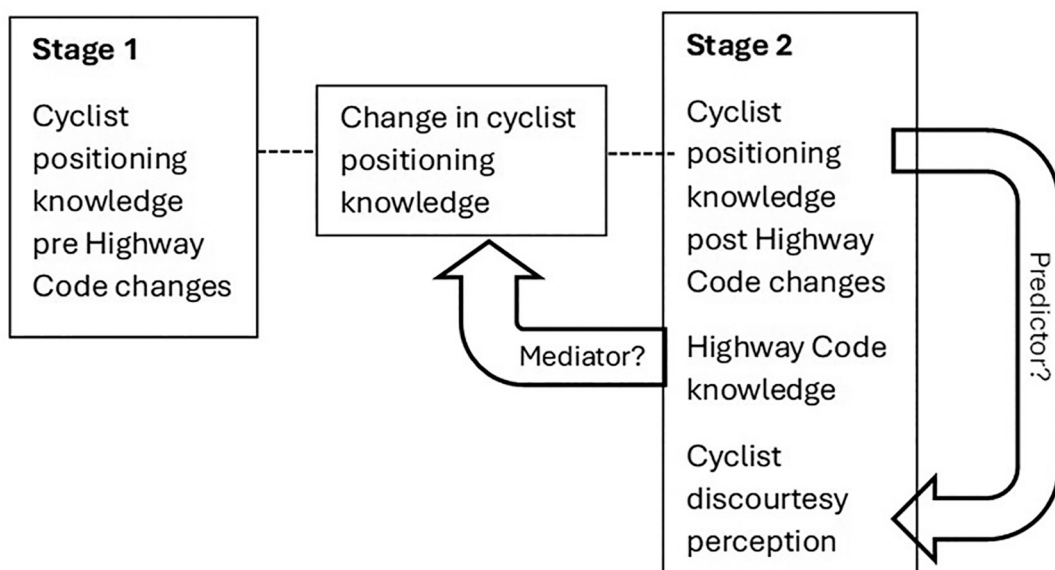


Fig. 3. Analysis block diagram.

Table 4

Changes in cyclist position knowledge scores between survey 1 and survey 2.

<i>Cyclist position knowledge</i>			
	Survey 1 (Aug 2020) All participants	Survey 1 (Aug 2020) Also completed survey 2	Survey 2 (Dec 2023-Feb 2024)
Participants	n = 421	n = 167	n = 167
A (tight to kerb)	52.2 %	53.1 %	38.7 %
B (secondary)	42.5 %	42.6 %	52.6 %
C (primary)	5.3 %	4.2 %	8.7 %

bootstrapped hierarchical multiple regression, summarised in Table 6. Variance inflation factors ranged between 1.033 and 2.129 indicating multicollinearity did not affect the model.

Both models significantly predicted the DV of cyclist discourtesy; $F(6, 148) = 6.131$, $p < 0.001$, $R^2 = 0.199$, and $F(7, 147) = 6.118$, $p < 0.001$, $R^2 = 0.226$ (see Table 6). Three of the five variables added significantly to the prediction; Years licence held, ($\beta = 0.273$, $t = 3.640$, $p < 0.001$), Frequency cycling ($\beta = -0.249$, $t = -2.360$, $p = 0.020$), and stage 2 positioning score ($\beta = -0.175$, $t = -2.244$, $p = 0.026$). Highway Code knowledge did not make a significant contribution, indicating that efforts to improve knowledge of cyclist positioning through publicised updates to the highway code have not translated to lower perceptions of cyclist discourtesy.

The change in R^2 between the two models was significant ($\Delta R^2 = 0.027$, $F_{\text{change}} = 0.026$), indicating that when controlling for participation in cycling and driving experience, people with greater knowledge of cyclist positioning practice tend to have lower perceptions of cyclist discourtesy.

3.2. Results for RQ2 “Has the level of knowledge around cyclist positioning improved since the January 2022 Highway Code update, and can any change be attributed to the highway code changes?”

A t -test showed a statistically significant change in the cyclist positioning scores between the two stages of the survey ($t(166) = -6.445$, $p < 0.001$).

The coefficients in Table 5 indicate a significant positive relationship between cyclist positioning score on the second survey and Highway Code knowledge ($r = 0.328$, $p < 0.01$). A smaller but still significant positive relationship was present when looking at how much scores improved between the two surveys and Highway Code knowledge ($r = 0.165$, $p < 0.05$). There was also a positive correlation between Highway Code knowledge and cyclist positioning scores on the first survey ($r = 0.171$, $p < 0.05$) that took place before the highway code was updated.

A bootstrapped hierarchical multiple regression was carried out to determine the effect of highway code knowledge on the changes between the two surveys in cyclist positional knowledge, while controlling for potentially confounding variables. An ANOVA approach was used as this method is robust to non-normally distributed data when the group sizes are equal (Field, 2009, p.360). Table 7 shows this model was not significant; $F(7, 147) = 1.178$, $p = 0.319$, $R^2 = 0.053$.

As neither model is significant it demonstrates that the highway code knowledge variable does not predict changes in cyclist road positioning score.

4. Discussion

4.1. Key findings

It has been previously established that negative attitudes and behaviours towards cyclists by some drivers is a major barrier preventing many people from using bicycles for transport, restricting the wide-ranging benefits brought to society by a move from motorised to pedal powered transport. The aim of this study was to determine whether greater knowledge of recommended cyclist road positioning practice can predict more favourable interpretations of cyclist behaviour among drivers. It also examined whether changes to the Highway Code, applied and publicised in January 2022, can be linked to any improvement in understanding of cyclist positioning.

In line with our hypothesis, greater knowledge of cyclist positioning practice predicts lower perceptions of cyclist discourtesy. This model controls for potentially confounding factors that may affect whether an individual is positively or negatively disposed towards cyclists, comprising how much their lifestyle involves driving, years licence held, how they identify as a road user (driver / cyclist) and how frequently they cycle on the road themselves. This finding implies that increasing driver knowledge of cyclist positioning may enable a more positive interpretation of observed cyclist behaviour which could lead to a more favourable opinion of cyclists overall. A more favourable opinion of cyclists among drivers could help create a less hostile environment for people to travel by bicycle.

Within the cohort that took part in both stages of this study there was a significant improvement between 2020 and 2023 in scores for knowledge of appropriate road positioning for cyclists. Pearson correlations showed that Highway Code knowledge and recollection were significantly correlated with the improvement in positioning scores between the two iterations of the survey. However, a bootstrapped hierarchical multiple regression model using change in cyclist positioning score as the dependent variable, and Highway Code knowledge as the additional second stage independent variable, was not significant. This showed that highway code knowledge could not predict the change in cyclist positioning score when controlling for Frequency cycling, Highway Code recollection, Driving

Table 5
Pearson correlations.

		Mean	SD	Pearson Correlations											
				1	2	3	4	5	6	7	8	9	10	11	12
1	Cyclist position score change	5.52	11.692	—											
2	Highway Code knowledge	0.66	0.915	0.165*	—										
3	Sex	1.51	0.539	0.038	−0.073	—									
4	Age	39.56	13.718	0.038	0.107	−0.014	—								
5	Years licence held	19.55	13.646	0.055	0.085	−0.014	0.948**	—							
6	RUI (cyclist:driver)	12.4	21.66	−0.013	−0.004	0.177*	0.006	−0.038	—						
7	Driving Purposes	3.08	1.743	0.030	−0.027	0.033	−0.099	−0.034	−0.159*	—					
8	Highway Code change recall	0.5	0.502	0.154*	0.126	0.089	0.263**	0.237**	0.061	−0.003	—				
9	Cyclist position score stage2	23.7	11.896	0.517**	0.328**	0.008	0.091	0.094	0.101	−0.075	0.147	—			
10	Frequency cycling	0.55	0.955	−0.039	−0.021	0.245**	−0.038	−0.052	0.685**	0.052	0.042	0.055	—		
11	Cyclist Discourtesy	12.45	3.665	0.002	0.037	−0.131	0.275**	0.279**	−0.272**	0.012	0.077	−0.125	−0.308**	—	
12	Cyclist position score stage1	18.18	11.359	−0.476**	0.171*	−0.030	0.055	0.041	0.117	−0.107	−0.005	0.507**	0.096	−0.131	—

*. Correlation is significant at the 0.05 level (2-tailed).
**. Correlation is significant at the 0.01 level (2-tailed).

Table 6
Hierarchical multiple regression model for H1.

Model Summary ^c						
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate		
1	0.446 ^a	0.199	0.167	3.346		
2	0.475 ^b	0.226	0.189	3.301		
a. Predictors: (Constant), Frequency cycling, HC recall, Driving Purposes, HC knowledge, Years licence held, RUI						
b. Predictors: (Constant), Frequency cycling, HC recall, Driving Purposes, HC knowledge, Years licence held, RUI, stage2 pos score						
c. Dependent Variable: Cyclist Discourtesy						
ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	411.740	6	68.623	6.131	0<.001 ^b
	Residual	1656.647	148	11.194		
	Total	2068.387	154			
2	Regression	466.618	7	66.660	6.118	0<.001 ^c
	Residual	1601.769	147	10.896		
	Total	2068.387	154			
a. Dependent Variable: Cyclist Discourtesy						
b. Predictors: (Constant), Frequency cycling, HC recall, Driving Purposes, HC knowledge, Years licence held, RUI						
c. Predictors: (Constant), Frequency cycling, HC recall, Driving Purposes, HC knowledge, Years licence held, RUI, stage2 pos score						

Table 7
Multiple regression model for H2.

Model Summary ^c						
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate		
1	0.163 ^a	0.027	−0.013	11.767		
2	0.230 ^b	0.053	0.008	11.645		
<i>a. Predictors: (Constant), Frequency cycling, HC recall, Driving Purposes, Years licence held, Cyclist Discourtesy, RUI</i>						
<i>b. Predictors: (Constant), Frequency cycling, HC recall, Driving Purposes, Years licence held, Cyclist Discourtesy, RUI, HC knowledge</i>						
<i>c. Dependent Variable: pos score change</i>						
ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	561.204	6	93.534	0.676	0.670 ^b
	Residual	20491.51	148	138.456		
	Total	21052.71	154			
2	Regression	1117.854	7	159.693	1.178	0.319 ^c
	Residual	19934.86	147	135.611		
	Total	21052.71	154			
<i>a. Dependent Variable: pos score change</i>						
<i>b. Predictors: (Constant), Frequency cycling, HC recall, Driving Purposes, Years licence held, Cyclist Discourtesy, RUI</i>						
<i>c. Predictors: (Constant), Frequency cycling, HC recall, Driving Purposes, Years licence held, Cyclist Discourtesy, RUI, HC knowledge</i>						

Purposes, years licence held, perceived cyclist discourtesy and road user identity (cyclist / driver). The fact the Highway Code knowledge did not predict scores in selecting the appropriate position for cyclists suggests this knowledge alone has limited or negligible relevance, with drivers needing a more practical understanding of the guidance to understand how it applies to on-road scenarios. Some of this practical understanding may have been evident in the second stage of the survey as it took place after the COVID-19 lockdowns. The lockdowns would have given participants the opportunity to get used to the elevated numbers of cyclists on the roads (Department for Transport, 2021), or have spent some time themselves cycling on the road – building first-hand experience of how cyclists interact with their environment.

The variables included in the model do not explain the reason for the improved cyclist positioning scores, but the improvement in scores relating to perceptions of cyclist discourtesy indicate the impact of this improvement could support reduced hostility towards cyclists. The data examined for this study indicates that simply being a few years older and more experienced is unlikely to be an explanation for this improvement because years licence held did not contribute significantly to the model.

One reason for the improvement in scores could be an indirect effect of the Highway Code being disseminated via word-of-mouth where the precise rules are lost but the general spirit of the guidance is retained. Conflict between cyclists and motorists is a prominent current affairs issue in the UK, for example featuring as a half hour special on the BBC prime time current affairs programme Panorama in November 2022 ('Road Rage: Cars v Bikes', 2022). This creates much discourse around the issue giving opportunities for information to spread. This could include information around cyclist positioning practice being shared independently of the Highway Code changes. Another possibility is that the participants may have subconsciously picked up on the changes more easily, having previously answered questions on the topic in stage 1 of the survey.

4.2. Limitations

The mean age of participants in this study was 39.56, over 10 years younger than the average licence holder (Department for Transport, 2024) and almost 10 years younger than the average for the eligible population aged 18+ (Nomis, 2023). These variables were controlled for in the analysis but this may limit how well the results can be applied to the wider population.

Highway code knowledge was measured using a multiple-choice question and a graphical sliding scale. The options given in the questions may assist participants in making an educated guess as to the correct answer. This may lead to an overestimate of actual highway code knowledge among the sample.

4.3. Further work

The current study found that knowledge of cyclist positioning was a significant component in variations in perceptions of cyclist discourtesy. Further work is needed to understand how best to drive increases in knowledge of correct cyclist positioning.

Published literature discussed in this paper supports the legitimisation of cyclists as a measure that has potential to create more courtesy shown towards cyclists (Rissel et al., 2002, Oldmeadow et al., 2019; Johnson et al., 2014). Further work is needed pilot and evaluate an educational intervention that uses both factual information and techniques such as humanising cyclists to help non cyclists understand the legitimacy of cycling in accordance with the Highway Code.

The link between specific knowledge around cyclist positioning and specific appraisals of situations where cyclists are legitimately taking a centre lane position requires exploration. Further work is needed to understand specific scenarios that are perceived as examples of cyclist discourtesy. This could inform a knowledge-based intervention that focuses on where people cycling in accordance with the highway code are perceived as discourteous. This could work by addressing specific misconceptions about the reasons for certain cyclist behaviours that drivers find frustrating, by explaining why these behaviours are legitimate and rational ways to safely share the road with others, and not examples of cyclists being selfish, arrogant or entitled.

Recent research into humanising cyclists (Crundall and Van Loon, 2023) has shown promising results with more positive explicit attitudes and more cautious intended overtaking behaviour after watching a video designed to humanise a cyclist. An intervention that combines this humanising approach, with a demonstration of the credibility given to cyclist road positioning by the highway code, may be even more effective.

Not all drivers have a hostile disposition towards cyclists. Further work may focus on drivers who have been pre-selected for having a hostile disposition towards cyclists. Studying such a group of drivers may shed light on underlying reasons for this hostile disposition, including misconceptions around correct cyclist positioning, the impact of incidents where people on bicycles were not riding in accordance with the highway code, and rhetoric in the media and online relating to cycling.

5. Conclusions

The second stage of a hierarchical multiple regression model significantly predicted the dependent variable of cyclist discourtesy $F(7, 147) = 6.118, p < 0.001, R^2 = 0.226$, with the measure of applied cyclist lane positioning knowledge adding significantly to this prediction ($\beta = -0.175, t = -2.244, p = 0.026$) and accounting for a significant change compared to the first stage model that excluded lane position knowledge ($\Delta R^2 = 0.027, F_{\text{change}} = 0.026$). These results support hypothesis 1, as they show cyclist lane positioning knowledge predicts lower perceived cyclist discourtesy when controlling for participation in cycling and driving experience. Hypothesis 2 was not supported as although the results showed there has been a significant improvement in the level of applied cyclist lane positioning knowledge within the group across the two stages of the survey ($t(166) = -6.445, p < 0.001$), a bootstrapped hierarchical multiple regression showed the variable of highway code knowledge was not a significant predictor of lane positioning knowledge ($F(7, 147) = 1.178, p = 0.319, R^2 = 0.053$).

This study has shown drivers who are better able to select suitable road positions for cyclists have lower perceptions of cyclist discourtesy and this effect remains significant when controlling for potential in group bias resulting from the driver's own participation in cycling. There were significant improvements in participant's ability select suitable road positions for cyclists between 2020 and 2023, but these improvements could not be linked to specific knowledge of the Highway Code changes that formalised cyclist road positioning advice in 2022.

Our study provides support for exploring the benefits of improving driver knowledge of how cyclists interact with their environment. Greater knowledge of cyclist positioning has the potential to re-frame specific interactions in a way that leads drivers towards a less negative interpretation of the cyclist's behaviour. If these interactions are consistently positive, then over time they could lead to a more positive underlying attitude. Improving applied knowledge of how to select appropriate cyclist road positions (but not knowledge of the specific rules) significantly predicts lower perceptions of cyclist discourtesy. This could help reduce driver hostility towards cyclists, making cycling an attractive transport option for more people. This research provides the theoretical basis for creating and evaluating cycling-specific knowledge-based interventions for a target audience of motor vehicle drivers that has the potential to reduce risky overtaking manoeuvres and aggression directed towards cyclists, making cycling a more attractive transport option to people who are currently discouraged from doing so because of the perceived danger posed by motor vehicles.

CRedit authorship contribution statement

William Cubbin: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis,

Data curation, Conceptualization. **Kjell van Paridon**: Writing – review & editing, Supervision, Methodology, Conceptualization. **Matthew Timmis**: Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Helen Keyes**: Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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